

Swaroop S. Hegde

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Research Interests

Additive Combinatorics, Ramsey Theory, Probabilistic Methods in Combinatorics

Education

University of Georgia 2021 - present
 PhD in Mathematics Athens, GA
 Advisor: Giorgis Petridis

National Institute of Science Education and Research (NISER) 2015 - 2020
 Integrated M.Sc. in Mathematics Bhubaneswar, India
 Advisor: Brundaban Sahu

- Master's Thesis: Combinatorics and Representation Theory of Symmetric Groups

Work Experience

Instructor of Record, University of Georgia Aug 1, 2021 - present
 Responsible for most aspects of the following courses including preparing course notes, delivering lectures, writing exams and quizzes, holding office hours and managing grades.

- Calculus I (Math 2250): Spring 2025, Fall 2024, Fall 2023, Spring 2023, Fall 2022
- PreCalculus (Math 1113): Fall 2026 (x2), Spring 2026 (x2), Fall 2025 (x2), Spring 2024 (x2), Spring 2022 (x2), Fall 2021
- Online Precalculus (Math 1113E): Summer 2024

Publications

A somewhat sure note on an un-Schur problem Sep 2026
 S. Hegde, H. Kumar, Pratibha
 arXiv preprint: [arXiv:2609.18474](https://arxiv.org/abs/2609.18474)

Refined upper bounds on Schur-like numbers Aug 2026
 S. Hegde, A. Lott, G. Petridis, N. R. Ponagandla
 arXiv preprint: [arXiv:2608.03661](https://arxiv.org/abs/2608.03661)

Extensions of the Furstenberg-Sárközy theorem via the arithmetic level-d inequality May 2026
 C. F. E. Adajar, R. Agrawal, M. R. Choudhuri, C. Y. Chuah, S. Fan, S. Hegde, A. Lott, K. Nandakumar, and N. R. Ponagandla
 arXiv preprint: [arXiv:2605.16216](https://arxiv.org/abs/2605.16216)

An Inverse Result for Ruzsa's Inequality on Triple Sumsets Oct 2025
 S. Hegde
 arXiv preprint: [arXiv:2510.20073](https://arxiv.org/abs/2510.20073)

Awards

David Galewski Outstanding Graduate Teaching Award Apr 2024

Mathematics Department, University of Georgia

Outstanding Teaching Assistant Award

Mar 2024

Center for Teaching and Learning, University of Georgia

Certificate of Merit

Sep 2018

School of Mathematical Sciences, NISER — For outstanding performance in the Spring 2018 semester.

INSPIRE Scholarship Award (2015-2020)

Aug 2015

Department of Science and Technology, Govt. of India

Selected Talks

Generalized Schur Numbers

Sep 2026

UGA Analysis Seminar

An Inverse Result for Ruzsa's Inequality on Triple Sumsets

Mar 2026

Graduate Student Combinatorics Conference, University of Illinois at Chicago

An Inverse Result for Ruzsa's Inequality on Triple Sumsets

Oct 2025

Discrete Math Seminar, Iowa State University (virtual)

An Inverse Result for Ruzsa's Inequality on Triple Sumsets

May 2025

CanADAM Conference, Ottawa, Canada

An Inverse Result for Ruzsa's Inequality on Triple Sumsets

May 2025

Integers Conference, UGA

Ramsey number $R(4,k)$

Mar 2024

Oral Qualifying Exam Presentation, UGA

AKS Algorithm for Primality Testing

Oct 2023

Combinatorial Number Theory Reading Seminar, UGA

A Polygon Problem

Oct 2023

Graduate Student Seminar, UGA

The RSK Correspondence

Apr 2023

Graduate Student Seminar, UGA

Almost Periods of Convolutions and Application to Roth's Theorem

Feb 2023

Additive Combinatorics Reading Seminar, UGA

Cores and Quotients of Integer Partitions

Nov 2022

Graduate Student Seminar, UGA

Rédei's Theorem and Szőnyi Method

Sep 2021

Polynomial Methods Reading Seminar, UGA

Roth's Theorem via Finite Fourier Analysis

Apr 2019

Math Seminar, NISER

Pebbling the Chessboard

Apr 2018

NISER Open Day

The Story of Dimension

Feb 2017

MathematiX Club, NISER

Clock Arithmetic

Apr 2016

NISER Open Day (for high school students)

Gödel's Incompleteness Theorems

Nov 2015

MathematiX Club, NISER

Conferences and Workshops

Graduate Workshop in Logic and Combinatorics

Aug 2026

University of Illinois at Chicago

International Conference on Graph Theory and its Applications (ICGTA 2026)

Jun 2026

Presidency University, Bangalore, India

Simons School and Workshop on Harmonic Analysis and applications to Ramsey theory

Jun 2026

Rényi Institute, Budapest, Hungary

Simons School and Workshop on Harmonic Analysis and Analytic Number Theory

May 2026

Rényi Institute, Budapest, Hungary

Graduate Student Combinatorics Conference

Mar 2026

University of Illinois at Chicago

PCMI 2025 Graduate Summer School on Extremal Combinatorics and Probabilistic Methods

Jul 2025

PCMI, Park City, Utah

Graduate Research Workshop in Combinatorics

Jun 2025

Iowa State University, Ames, Iowa

CanaDAM Conference 2025

May 2025

Ottawa, Ontario, Canada

Integers Conference 2025

May 2025

University of Georgia, Athens, GA

Algebraic and Analytic Methods in Combinatorics

Mar 2025

SLMath, Berkeley, California

Introductory Workshop - Graph Theory: Extremal, Probabilistic and Structural

Feb 2025

SLMath, Berkeley, California

Early-Career Conference in Combinatorics

Jun 2024

University of Illinois Urbana-Champaign, IL

33rd Cumberland Conference on Combinatorics, Graph Theory & Computing

May 2024

Mississippi State University, MS

Formalization of Mathematics

Jun 2023

SLMath, Berkeley, California

Integers Conference 2023

May 2023

University of Georgia, Athens, GA

ATM-Schools AIS on Representation Theory of Finite Groups

Jun 2017

Chennai Mathematical Institute, Chennai, India

Madhava Mathematics Camp

Jun 2017

Indian Statistical Institute, Bangalore, India

National Science Camp (VIJYOSHI)

Dec 2015

Kolkata, India

Selected Coursework

Selected Graduate Courses: Entropy Methods in Combinatorics, Ramsey Theory, Additive Combinatorics I & II, Probabilistic Methods in Combinatorics, Ergodic Theory, The Circle Method, Equations with Prime Variables, Algebraic Graph Theory, Quiver Representations, Commutative Algebra, Lie Algebras, Hopf Algebras

Selected Reading Courses: Asymptotics of Ramsey Number $R(4,t)$ (Oral Exam), Kelley-Meka bound for Roth's theorem, Some recent improvements on Freiman's Theorem, Polynomial Methods in Combinatorics, Reflection Groups

Mentoring

Directed Reading Program, UGA Math Department

Jun 2021 - present

Mentored UGA undergraduate students on various projects

- Yan Heng: Formalization of Mathematics in Lean, Spring 2026
- Benjamin Wang-Tie: Applications of Entropy in Combinatorics, Spring 2025
- Aakash Khanal: Mathematics of Card Shuffling, Summer 2021

Service

Volunteer for Sonia Kovalevsky Day

Apr 25, 2026

University of Georgia

Co-organizer of the Graduate Student Seminar

Jan 1, 2025 - May 1, 2025

Mathematics Department, University of Georgia

Co-facilitator of the Precalculus community working group

Jan 1, 2024 - May 1, 2024

Mathematics Department, University of Georgia

Volunteer for High School Math Tournament

Oct 1, 2023 - Oct 1, 2024

University of Georgia — Volunteered in Oct 2023 and Oct 2024.

Peer Mentor for first year graduate students

Aug 1, 2023 - May 1, 2025

Mathematics Department, University of Georgia

Secretary of Student Chapter of AMS

Aug 1, 2022 - May 1, 2023

Mathematics Department, University of Georgia